Problem Set 10

Grant Talbert
April 24, 2025
PY 451 Section A1

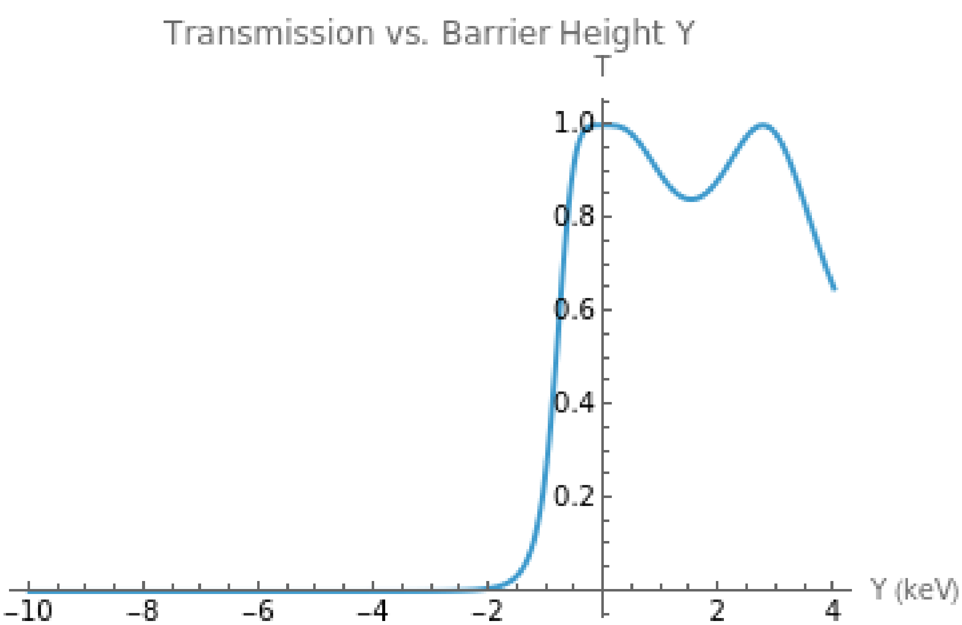
Boston University

1 Scattering

1.

```
haiiii:=1.60218*10^(-16) (* conversion keV -> J *)
e := 10^(-18) * haiiii (* keV *)
m := 9.1093837139*10^(-31) (*kg*)
hbar:=1.054571817*10^(-34)  (*J s*)
a:=0.01 (*m*)

V0[Y_]:=10^(-18)*Y*haiiii

k1:=Sqrt[ (2*m*e)/(hbar^2)]
k2[Y_]:=Sqrt[(2*m*(e+V0[Y]))/(hbar^2)]

T[Y_]:=1/(1+(((k1^2-k2[Y]^2)^2*(Sin[2*k2[Y]*a])^2)/(4*k1^2*k2[Y]^2)))
Plot[
  T[Y],
  {Y, -10, 4},
  PlotRange -> All,
  AxesLabel -> {"Y (keV)", "T"},
  PlotLabel  -> "Transmission vs. Barrier Height Y"
]

```

Output: The equation peaks at 1 twice, around $Y = 0$ and $Y = 2.3$. It's also basically 0 until



$Y = -2$, at which point it shoots up. After attaining it's final peak, it goes down.

2. No thanks

2 3d Wavefunctions

1. Note - when I switched all the integrals into cylindrical coordinates, I used θ as the variable instead of φ because I forgot we were using θ as the step function. Because of this, all of my integrals use θ as a function AND as the variable to be integrated on. I can't be asked fixing the notation, and either way it should be clear when θ is the step function and when it is the variable. Because $\left|e^{i\theta}\right| = 1$, and since we clearly have $|\theta(x)|^2 = \theta(x)$ since θ can only equal 0 or 1, we have

$$
\begin{aligned}
\langle\psi\,|\,\psi\rangle &= \iiint\limits_{\mathbb{R}^3} A^2\left|e^{-z^2/a^2}\theta\big(b^2-\big(x^2+y^2\big)\big)\right|^2 \mathrm{d}\mu \\
&= A^2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\theta\big(b^2-\big(x^2+y^2\big)\big)\int_{-\infty}^{\infty}\Big(e^{-z^2/a^2}\Big)^2\,\mathrm{d}z\mathrm{d}y\mathrm{d}x \\
&= A^2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\theta\big(b^2-\big(x^2+y^2\big)\big)\int_{-\infty}^{\infty}e^{-z^2\left(2/a^2\right)}\,\mathrm{d}z\mathrm{d}y\mathrm{d}x \\
&\text{Let } u = z\frac{\sqrt{2}}{a} \\
&= A^2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\theta\big(b^2-\big(x^2+y^2\big)\big)\int_{-\infty}^{\infty}\frac{a}{\sqrt{2}}e^{-u^2}\,\mathrm{d}u\mathrm{d}y\mathrm{d}x \\
&= A^2a\sqrt{\frac{\pi}{2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\theta\big(b^2-\big(x^2+y^2\big)\big)\,\mathrm{d}y\mathrm{d}x \\
&= A^2a\sqrt{\frac{\pi}{2}}\int_{0}^{2\pi}\int_{0}^{\infty}\theta\big(b^2-r^2\big)r\,\mathrm{d}r\mathrm{d}\theta \\
&= A^2a\sqrt{\frac{\pi}{2}}\int_{0}^{2\pi}\int_{0}^{|b|}r\,\mathrm{d}r\mathrm{d}\theta + A^2a\sqrt{\frac{\pi}{2}}\int_{0}^{2\pi}\int_{|b|}^{\infty}0\,\mathrm{d}r\mathrm{d}\theta \\
&= A^2a\sqrt{\frac{\pi}{2}}\pi b^2 \\
&= \frac{A^2ab^2\sqrt{\pi^3}}{\sqrt{2}} \\
&\equiv 1,
\end{aligned}
$$

where I have set $\theta^2 = \theta$ due to the fact that

$$
\theta\big(b^2-r^2\big) = \begin{cases} 1, & |b| \geq |r| \\ 0, & |b| < |r| \end{cases}.
$$

It follows that

$$
A = \sqrt{\frac{\sqrt{2}}{ab^2\sqrt{\pi^3}}}.
$$

2. First, we compute the partial derivatives

$$
\begin{aligned}
\frac{\partial\psi}{\partial x} &= Ae^{-z^2/a^2}e^{i(p_x x+p_y y)/\hbar}\left(\frac{ip_x}{\hbar}\theta\big(b^2-\big(x^2+y^2\big)\big) - 2x\delta\big(b^2-\big(x^2+y^2\big)\big)\right), \\
\frac{\partial\psi}{\partial y} &= Ae^{-z^2/a^2}e^{i(p_x x+p_y y)/\hbar}\left(\frac{ip_y}{\hbar}\theta\big(b^2-\big(x^2+y^2\big)\big) - 2y\delta\big(b^2-\big(x^2+y^2\big)\big)\right).
\end{aligned}
$$

For simplicity, denote by ϕ the exponential stuff outside of the parenthases. The notation $\phi(x)$ should be interpreted as "$\phi \times x$", not as inserting x into the function ϕ. In other words, just interpret it as a constant, as I will expand it out any time we actually need to work with the stuff inside of it. We then have

$$\partial_x \psi = \phi\left(\frac{ip_x}{\hbar}\theta - 2x\theta'\right), \qquad \partial_y \psi = \phi\left(\frac{ip_y}{\hbar}\theta - 2y\theta'\right),$$

with the implied arguments for the function θ. We then have

$$(XP_y - YP_x)\psi = (-i\hbar x\partial_y + i\hbar y\partial_x)\psi = i\hbar\phi\left(2xy\theta' - 2xy\theta' - \frac{ixp_y}{\hbar}\theta + \frac{iyp_x}{\hbar}\theta\right) = \phi(xp_y - yp_x)\theta.$$

Therefore, $\langle\psi|(XP_y - YP_x)|\psi\rangle$ equals the following.

$$\begin{aligned}
&\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \phi^2\theta\big(b^2 - (x^2 + y^2)\big)(xp_y - yp_x)\theta\big(b^2 - (x^2 + y^2)\big)\,\mathrm{d}z\mathrm{d}y\mathrm{d}x \\
&= A^2\int_0^{2\pi}\int_0^{\infty}\int_{-\infty}^{\infty} e^{-z^2(2/a^2)}\theta\big(b^2 - r^2\big)(p_y r\cos\theta - p_x r\sin\theta)r\,\mathrm{d}z\mathrm{d}r\mathrm{d}\theta \\
&= A^2 a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\int_0^{\infty} r^2\theta\big(b^2 - r^2\big)(p_y\cos\theta - p_x\sin\theta)\,\mathrm{d}r\mathrm{d}\theta \\
&= A^2 a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\int_0^{|b|} r^2(p_y\cos\theta - p_x\sin\theta)\,\mathrm{d}r\mathrm{d}\theta \\
&= A^2 a\sqrt{\frac{\pi}{2}}\int_0^{2\pi} \frac{|b|^3}{3}(p_y\cos\theta - p_x\sin\theta)\,\mathrm{d}\theta \\
&= \frac{A^2 a|b|^3}{3}\sqrt{\frac{\pi}{2}}p_y(1-1) \\
&= 0.
\end{aligned}$$

3. Reusing our result from part 2, we have

$$(XP_y - YP_x)\psi = \phi(xp_y - yp_x)\theta,$$

once again with implied arguments for θ to simplify notation. We now want to find

$$(XP_y - YP_x)(\phi(xp_y - yp_x)\theta).$$

Note that

$$\frac{\partial}{\partial x}(xp_y - yp_x)\theta = p_y\theta - 2x(xp_y - yp_x)\theta'.$$

Also note that

$$\frac{\partial}{\partial x}\phi = \frac{ip_x}{\hbar}\phi.$$

We thus have

$$\frac{\partial}{\partial x}(\phi(xp_y - yp_x)\theta) = \frac{ip_x}{\hbar}\phi(xp_y - yp_x)\theta + \phi p_y\theta - 2x\phi(xp_y - yp_x)\theta'.$$

By analogous logic,

$$\frac{\partial}{\partial y}(\phi(xp_y - yp_x)\theta) = \frac{ip_y}{\hbar}\phi(xp_y - yp_x)\theta - \phi p_x\theta - 2y\phi(xp_y - yp_x)\theta'.$$

Multiplying through by y and x respectively, and subtracting, we obtain

$$\frac{ixp_y}{\hbar}\phi(xp_y - yp_x)\theta - \phi xp_x\theta - 2xy\phi(xp_y - yp_x)\theta' - \left(\frac{iyp_x}{\hbar}\phi(xp_y - yp_x)\theta + \phi yp_y\theta - 2xy\phi(xp_y - yp_x)\theta'\right)$$

$$= \left(\frac{i}{\hbar}\phi(xp_y - yp_x)(xp_y - yp_x) - \phi xp_x - \phi yp_y\right)\theta.$$

Multiplying through by $-i\hbar$ gives

$$(XP_y - YP_x)^2\psi = \big(\phi(xp_y - yp_x)^2 - i\hbar\phi(xp_x + yp_y)\big)\theta.$$

Calculating $\langle\psi|(XP_y - YP_z)^2|\psi\rangle$ gives

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} A^2 e^{-z^2(2/a^2)}\theta\big(b^2 - \big(x^2 + y^2\big)\big)\Big((xp_y - yp_x)^2 - i\hbar(xp_x + yp_y)\Big)\,\mathrm{d}z\mathrm{d}y\mathrm{d}x$$

$$= A^2a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\int_0^{\infty}\theta\big(b^2 - r^2\big)\big(r^2p_y^2\cos^2\theta + r^2p_x^2\sin^2\theta - 2rp_xp_y\cos\theta\sin\theta - ri\hbar(p_x\cos\theta + p_y\sin\theta)\big)r\,\mathrm{d}r\mathrm{d}\theta$$

$$= A^2a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\int_0^{|b|} r^3p_y^2\cos^2\theta + r^3p_x^2\sin^2\theta - 2r^2p_xp_y\cos\theta\sin\theta - r^2i\hbar(p_x\cos\theta + p_y\sin\theta)\,\mathrm{d}r\mathrm{d}\theta$$

$$= A^2a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\frac{b^4}{4}p_y^2\cos^2\theta + \frac{b^4}{4}p_x^2\sin^2\theta - 2\frac{|b|^3}{3}p_xp_y\cos\theta\sin\theta - \frac{|b|^3}{3}i\hbar(p_x\cos\theta + p_y\sin\theta)\,\mathrm{d}\theta$$

$$= A^2a\sqrt{\frac{\pi}{2}}\int_0^{2\pi}\frac{b^4p_y^2}{8}(\cos 2\theta + 1) + \frac{b^4p_x^2}{8}(1 - \cos 2\theta) - \frac{|b|^3p_xp_y}{3}\sin 2\theta - \frac{|b|^3i\hbar}{3}(p_x\cos\theta + p_y\sin\theta)\,\mathrm{d}\theta$$

$$= A^2a\sqrt{\frac{\pi}{2}}\left(-\frac{|b|^3i\hbar}{3}(p_x\cos\theta + p_y\sin\theta)_0^{2\pi} + \frac{1}{2}\int_0^{4\pi}\frac{b^4p_y^2}{8}(\cos\theta + 1) + \frac{b^4p_x^2}{8}(1 - \cos\theta) - \frac{|b|^3p_xp_y}{3}\sin\theta\,\mathrm{d}\theta\right)$$

$$= \frac{A^2a}{2}\sqrt{\frac{\pi}{2}}\int_0^{4\pi}\frac{b^4p_y^2}{8}\cos\theta - \frac{b^4p_x^2}{8}\cos\theta - \frac{|b|^3p_xp_y}{3}\sin\theta\,\mathrm{d}\theta + \frac{A^2a}{2}\sqrt{\frac{\pi}{2}}\left(\frac{b^4p_y^2\pi}{2} + \frac{b^4p_x^2\pi}{2}\right)$$

$$= \frac{A^2ab^4\pi}{4}\sqrt{\frac{\pi}{2}}\big(p_x^2 + p_y^2\big).$$